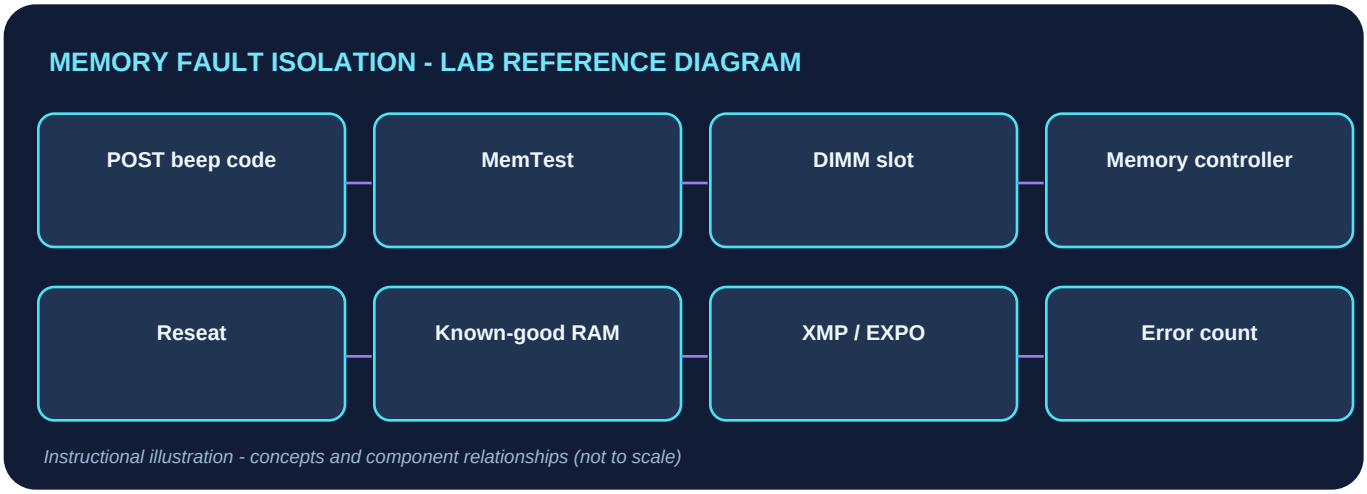
3.5 Troubleshoot Memory

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Diagnose POST and RAM issues.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Document error and power down safely.
2. Test with one known-good module.
3. Try supported slots one at a time.
4. Restore stable settings and verify with a memory test.

Troubleshooting and safe practice

• Intermittent freezes: run extended memory diagnostics.

• No POST: disable unstable memory profile and retest.

3.5 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
POST beep code	Firmware audio pattern indicating a detected startup issue.
MemTest	Observation or tool used to identify faults and confirm the repair.
DIMM slot	Physical interface or connection point; verify its intended function and compatibility before use.
Memory controller	Memory characteristic or configuration item; check motherboard support and installation requirements.
Reseat	A key term in memory fault isolation; identify its purpose, where it is used, and how to verify it.
Known-good RAM	Memory characteristic or configuration item; check motherboard support and installation requirements.
XMP / EXPO	A key term in memory fault isolation; identify its purpose, where it is used, and how to verify it.
Error count	Observation or tool used to identify faults and confirm the repair.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____